Semiclassical Consistency of the Weil Representation on $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$: Quantitative Sinc Embedding, Generator Convergence, and Aliasing Control

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2026

Abstract

We prove that the rescaled Weil generators $\hat{X}_q$ and $\hat{P}_q$ of the discrete Heisenberg group $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ converge strongly to the position operator x and the momentum operator $-i\partial_x$ as $q \to \infty$, in the sense of Hypothesis [H2] of the Cosmochrony $Q5a$ programme [2]. The proof rests on two structural lemmas. First, a discrete Sobolev estimate controls the error between the finite-difference operator and the continuous derivative at the ℓ^2 level. Second, a quantitative Poisson aliasing lemma controls the transition from $\ell^2(\mathbb{Z}/q\mathbb{Z})$ to $L^2(\mathbb{R})$ with an explicit rate $O(q^{-1})$ for data in $\mathcal{S}(\mathbb{R})$. As a consequence we obtain a quasi-isometry lemma for the sinc embedding ι_q with the explicit rate

$$\left| \|\iota_q f\|_{L^2}^2 - \|f\|_{\ell^2}^2 \right| \leq \frac{C}{q} \mathcal{N}(\psi),$$

which is a reusable structural estimate for the identification of the effective metric constant A in terms of the Born–Infeld saturation constant c_{BI} (open problem $Q5a$-O5 and [8]). The proof of [H2] closes the last open hypothesis of the $Q5a$ large-q limit theorem, completing the geometric emergence sub-programme of [5, 4, 7, 8].

Keywords. Weil representation; Heisenberg group; semiclassical limit; sinc embedding; discrete Sobolev estimate; aliasing control; Schwartz convergence; Cosmochrony; $Q5a$.

Contents

1 Introduction **3**
1.1 Context and motivation 3
1.2 Structure of the argument 3
1.3 Relation to previous work 4
1.4 Reusability of the quasi-isometry lemma 4
1.5 Organisation 4

2 Sinc embedding and normalisations **5**
2.1 The sinc embedding 5
2.2 Weil generators and rescaled generators 5
2.3 Uniform operator bound 5

3 Discrete Sobolev estimate **7**

4 Uniform aliasing control **8**
4.1 Poisson aliasing lemma 8
4.2 Quasi-isometry with explicit rate 8

5 Convergence of the position generator **10**

6 Convergence of the momentum generator **11**

7 Strong convergence on Schwartz data **12**

8 Extension to $L^2(\mathbb{R})$ **13**

1 Introduction

1.1 Context and motivation

The Cosmochrony programme derives physical observables from the spectral structure of admissible non-injective projections on discrete Heisenberg groups $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ [1, 3]. The large-q limit of this structure is the subject of the $Q5a$ paper [2], which identifies the limiting Hilbert space as $L^2(\mathbb{R})$ and the limiting operators as the standard generators of the Schrödinger (metaplectic) representation of $\mathrm{Heis}_3(\mathbb{R})$. The main theorem of [2] (Theorem 7.1) is conditional on several hypotheses, of which all but one have been established by the geometric emergence sub-programme [5, 4, 7, 8, 6]. The sole remaining open hypothesis is [H2] (Hypothesis 4.3 of [2]):

Hypothesis 1.1 ([H2], [2])**.** *For every $\psi \in \mathcal{S}(\mathbb{R})$ and every $\varepsilon > 0$, there exists q_0 such that for all primes $q \geq q_0$,*

$$\left\| \iota_q\big(\hat{X}_q\, \iota_q^* \psi\big) - x\psi \right\|_{L^2(\mathbb{R})} < \varepsilon,$$
$$\left\| \iota_q\big(\hat{P}_q\, \iota_q^* \psi\big) - (-i\partial_x \psi) \right\|_{L^2(\mathbb{R})} < \varepsilon.$$

The convergence holds uniformly for ψ in bounded subsets of $\mathcal{S}(\mathbb{R})$.

The present paper proves Hypothesis 1.1.

1.2 Structure of the argument

The two operators $\hat{X}_q$ and $\hat{P}_q$ have asymmetric proof structures. For the position generator $\hat{X}_q$, the argument is quasi-pointwise: the character $e^{2\pi ik/q}$ is approximated by $1 + 2\pi ik/q$ via a Taylor expansion, and the error is controlled by the Schwartz decay of $\iota_q^*\psi$. For the momentum generator $\hat{P}_q$, the argument is genuinely non-local: a discrete Sobolev estimate (Lemma 3.1) is required to control the finite-difference error at the ℓ^2 level before the aliasing lemma (Lemma 4.1) can be applied to pass to $L^2(\mathbb{R})$. The logical chain for $\hat{P}_q$ is therefore:

discrete Sobolev estimate $\longrightarrow$ aliasing lemma $\longrightarrow$ $L^2(\mathbb{R})$ convergence.

This asymmetry is reflected explicitly in the proof structure of Sections 5 and 6.

1.3 Relation to previous work

Hannay and Berry [1] study the semiclassical limit of quantised maps on the torus in the dual regime $\hbar = (2\pi q)^{-1} \to 0$; their limit shares the same discrete-to-continuous mechanism. Degli Esposti and Graffi [2] provide structural support for [H2] via an analysis of the semiclassical limit of maps quantised on $\mathbb{Z}/q\mathbb{Z}$, but their results do not cover $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ in the parameterisation of [1]. The present paper fills this gap.

1.4 Reusability of the quasi-isometry lemma

The quasi-isometry bound of Corollary 4.2 (Section 4) is not merely a technical step in the proof of [H2]. It quantifies the defect between $\|\iota_q f\|_{L^2}^2$ and $\|f\|_{\ell^2}^2$ with a constant that depends explicitly on Sobolev norms of ψ but is independent of q. This bound is the missing input for open problem $Q5a$-O5, which asks for the identification of the effective metric constant

$$A = \sum_{n=1}^{\infty} \sigma^*(n) \approx \frac{c_{\mathrm{BI}}^2}{C_{\mathrm{Heis}}}\, \zeta\big(\tfrac{\delta^*}{2}\big)$$

in terms of the Born–Infeld saturation constant c_{BI} (cf. [8], formula (9)). We therefore state Corollary 4.2 as a numbered result.

1.5 Organisation

Section 2 defines the sinc embedding and its adjoint, fixes normalisations, and records the uniform operator bound on the Weil generators (Lemma 2.4). Section 3 proves the discrete Sobolev estimate (Lemma 3.1). Section 4 proves the quantitative aliasing lemma (Lemma 4.1) and derives the quasi-isometry corollary (Corollary 4.2). Section 5 proves convergence of $\hat{X}_q$. Section 6 proves convergence of $\hat{P}_q$. Section 7 assembles the strong convergence theorem on $\mathcal{S}(\mathbb{R})$. Section 8 extends the result to $L^2(\mathbb{R})$ by density.

2 Sinc embedding and normalisations

2.1 The sinc embedding

Definition 2.1 (Sinc embedding)**.** *Let q be a positive integer. The* sinc embedding $\iota_q : \ell^2(\mathbb{Z}/q\mathbb{Z}) \to L^2(\mathbb{R})$ *is defined by*

$$(\iota_q f)(x) := q^{1/2} \sum_{k \in \mathbb{Z}/q\mathbb{Z}} f(k)\,\mathrm{sinc}(qx - k)\,, \qquad \mathrm{sinc}(t) := \frac{\sin(\pi t)}{\pi t}.$$

Definition 2.2 (Sampling operator)**.** *The* sampling operator $\iota_q^* : L^2(\mathbb{R}) \to \ell^2(\mathbb{Z}/q\mathbb{Z})$ *is the L^2-adjoint of ι_q, given by*

$$(\iota_q^* \psi)(k) := q^{1/2}\, \psi\Big(\tfrac{k}{q}\Big)\,, \qquad k \in \mathbb{Z}/q\mathbb{Z}.$$

Remark 2.3. *The factor $q^{1/2}$ in Definition 2.1 ensures that $\iota_q \iota_q^*$ is approximately the identity on the low-frequency sector of $L^2(\mathbb{R})$, consistently with the Nyquist–Shannon sampling theorem. The factor $q^{1/2}$ in Definition 2.2 is the adjoint normalisation.*

2.2 Weil generators and rescaled generators

The Weil generators of $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ acting on $\ell^2(\mathbb{Z}/q\mathbb{Z})$ are

$$(\rho_q(X)f)(k) := e^{2\pi i k/q} f(k),$$
$$(\rho_q(Y)f)(k) := f(k + 1 \bmod q).$$

The *rescaled generators* are defined as in [2]:

$$\hat{X}_q := \frac{q}{2\pi i}\big(\rho_q(X) - I\big), \qquad \hat{P}_q := \frac{q}{2\pi i}\big(\rho_q(Y) - I\big).$$

2.3 Uniform operator bound

The following lemma is used in Section 8 to extend the $\mathcal{S}(\mathbb{R})$ convergence result to $L^2(\mathbb{R})$ by density.

Lemma 2.4 (Uniform operator bound)**.** *For all primes q and both generators,*

$$\|\rho_q(X) - I\|_{\ell^2(\mathbb{Z}/q\mathbb{Z}) \to \ell^2(\mathbb{Z}/q\mathbb{Z})} \le 2, \qquad \|\rho_q(Y) - I\|_{\ell^2(\mathbb{Z}/q\mathbb{Z}) \to \ell^2(\mathbb{Z}/q\mathbb{Z})} \le 2.$$

Consequently,

$$\big\|\hat{X}_q\big\|_{\ell^2(\mathbb{Z}/q\mathbb{Z}) \to \ell^2(\mathbb{Z}/q\mathbb{Z})} \le \frac{q}{\pi}, \qquad \big\|\hat{P}_q\big\|_{\ell^2(\mathbb{Z}/q\mathbb{Z}) \to \ell^2(\mathbb{Z}/q\mathbb{Z})} \le \frac{q}{\pi}.$$

After transport by ι_q, the operators $\iota_q \hat{X}_q \iota_q^*$ and $\iota_q \hat{P}_q \iota_q^*$ are bounded on $L^2(\mathbb{R})$ uniformly in q on the range of ι_q^*. Although the $\ell^2(\mathbb{Z}/q\mathbb{Z})$-operator norms $\left\|\hat{X}_q\right\|$, $\left\|\hat{P}_q\right\|$ grow as $O(q)$, the action on sampled Schwartz data $\iota_q^*\psi$ is effectively controlled via the quasi-isometry (Corollary 4.2): the transported operators $\iota_q \hat{X}_q \iota_q^*$ and $\iota_q \hat{P}_q \iota_q^*$ remain bounded uniformly in q on bounded subsets of $\mathcal{S}(\mathbb{R})$, since the Schwartz decay of ψ absorbs the $O(q)$ growth.

Proof sketch. Both $\rho_q(X)$ and $\rho_q(Y)$ are unitary on $\ell^2(\mathbb{Z}/q\mathbb{Z})$, so $\|\rho_q(s) - I\| \leq 2$ for $s \in \{X, Y\}$. The bound on the rescaled generators follows from the definition and the triangle inequality. □

3 Discrete Sobolev estimate

Let T_1 denote the shift operator on $\ell^2(\mathbb{Z}/q\mathbb{Z})$: $(T_1 g)(k) := g(k+1 \bmod q)$, and let T_{-1} denote its inverse. Define the *forward discrete derivative* $D_q g := (q/2\pi i)(T_1 g - g)$, which coincides with $\hat{P}_q g$ by definition, and the *centered discrete derivative*

$$g'_{\text{disc}} := \frac{q}{4\pi i}(T_1 - T_{-1})g,$$

which is the standard centered finite-difference approximation of the derivative at step size $1/q$.

Lemma 3.1 (Discrete algebraic identity)**.** *For every $g : \mathbb{Z}/q\mathbb{Z} \to \mathbb{C}$, the following identity holds exactly:*

$$D_q g - g'_{\text{disc}} = \frac{q}{4\pi i}(T_1 - 2I + T_{-1})g.$$

In particular,

$$\|D_q g - g'_{\text{disc}}\|_{\ell^2} = \frac{q}{4\pi} \, \|(T_1 - 2I + T_{-1})g\|_{\ell^2} \, .$$

Proof sketch. Direct computation: $(T_1 - I) = \frac{1}{2}(T_1 - T_{-1}) + \frac{1}{2}(T_1 - 2I + T_{-1})$. Multiplying both sides by $q/(2\pi i)$ gives the identity. □

Lemma 3.2 (Passage to the continuous second derivative)**.** *For $g = \iota_q^* \psi$ with $\psi \in \mathcal{S}(\mathbb{R})$,*

$$\left\|(T_1 - 2I + T_{-1})\iota_q^* \psi\right\|_{\ell^2} \leq C' \, \left\|\partial_x^2 \psi\right\|_{L^2(\mathbb{R})} \, .$$

Proof sketch. This follows from the aliasing lemma (Lemma 4.1 below) applied to the sampled second finite difference. □

Remark 3.3. *Lemma 3.1 is purely ℓ^2-intrinsic and does not reference any continuous norm. The passage to $\|\partial_x^2 \psi\|_{L^2}$ in Lemma 3.2 requires the aliasing control of Section 4. This creates a logical dependency: Section 3 is independent of Section 4 at the lemma level, but the application to $g = \iota_q^* \psi$ (used in Section 6) requires both.*

4 Uniform aliasing control

4.1 Poisson aliasing lemma

Lemma 4.1 (Quantitative aliasing bound)**.** *For $\psi \in \mathcal{S}(\mathbb{R})$ and any $M \geq 2$, there exists $C_M > 0$ depending only on ψ and M such that*

$$\sum_{k \neq 0} \left| \hat{\psi}(\xi + kq) \right| \leq \frac{C_M}{q^M}$$

uniformly in $\xi \in \mathbb{R}$. Consequently, for the integral aliasing error,

$$\left| \int_{\mathbb{R}} \hat{\psi}(\xi + kq)\, \overline{\hat{\psi}(\xi)}\, d\xi \right| \leq \frac{C'_M}{q^M}$$

for each $k \neq 0$.

Proof sketch. Since $\psi \in \mathcal{S}(\mathbb{R})$, its Fourier transform satisfies $\left|\hat{\psi}(\xi)\right| \leq C_M\,(1+|\xi|)^{-M}$ for all M, uniformly over all $\xi \in \mathbb{R}$ (this polynomial decay is the key property that makes the bound uniform in ξ). For $k \neq 0$ and $q \geq 1$, $|\xi + kq| \geq |k|\,q - |\xi|$, so for $|\xi| \leq q/2$ one has $|\xi + kq| \geq (|k| - \frac{1}{2})q$. Summing over $k \neq 0$ and bounding $\sum_{k\neq 0} |k|^{-M} < \infty$ for $M > 1$ yields the first bound. The estimate extends to all $\xi \in \mathbb{R}$ by periodicity of the aliasing decomposition. The integral estimate follows by Cauchy–Schwarz. □

4.2 Quasi-isometry with explicit rate

Corollary 4.2 (Quasi-isometry with rate $O(q^{-1})$)**.** *For $\psi \in \mathcal{S}(\mathbb{R})$ and $f = \iota_q^* \psi$,*

$$\left| \|\iota_q f\|_{L^2}^2 - \|f\|_{\ell^2}^2 \right| \leq \frac{C}{q}\, \mathcal{N}_2(\psi),$$

where $\mathcal{N}_2(\psi) := \|x\psi\|_{L^2}^2 + \|\partial_x \psi\|_{L^2}^2$ and $C > 0$ is independent of q.

Proof sketch. The Parseval identity for the sinc embedding gives

$$\|\iota_q f\|_{L^2}^2 - \|f\|_{\ell^2}^2 = \sum_{k \neq 0} \int_{\mathbb{R}} \hat{\psi}(\xi + kq)\, \overline{\hat{\psi}(\xi)}\, d\xi.$$

Applying Lemma 4.1 with $M = 2$ and bounding the sum over $k \neq 0$ by a geometric series in q yields the $O(q^{-1})$ rate. The explicit dependence on $\mathcal{N}_2(\psi)$ comes from the $M = 2$ decay estimate applied to the $k = \pm 1$ dominant terms. □

Remark 4.3 (Reusability for $Q5a$-O5)**.** *The rate $O(q^{-1})$ in Corollary 4.2 is summable with respect to the ℓ^2-energy weights $\sigma^*(n)$ provided $\delta^*/2 > 1$, a condition satisfied with large margin in the $O25$ numerical campaign [8]. This bound is therefore the quantitative input required for the identification $A \approx c_{\mathrm{BI}}^2/C_{\mathrm{Heis}}$ in open problem $Q5a$-O5.*

5 Convergence of the position generator

Theorem 5.1 (Convergence of $\hat{X}_q$)**.** *For every $\psi \in \mathcal{S}(\mathbb{R})$,*

$$\left\| \iota_q\big(\hat{X}_q\, \iota_q^*\psi\big) - x\psi \right\|_{L^2(\mathbb{R})} = O\big(q^{-1}\big)\,,$$

uniformly on bounded subsets of $\mathcal{S}(\mathbb{R})$.

Proof sketch. Since $(\hat{X}_q \iota_q^*\psi)(k) = \frac{q}{2\pi i}(e^{2\pi i k/q} - 1)(\iota_q^*\psi)(k)$, the Taylor expansion $e^{2\pi i k/q} = 1 + \frac{2\pi i k}{q} + O((k/q)^2)$ gives

$$(\hat{X}_q \iota_q^*\psi)(k) = k \cdot (\iota_q^*\psi)(k) + E_q(k), \qquad |E_q(k)| \le \frac{Ck^2}{q}\left|(\iota_q^*\psi)(k)\right|.$$

Step 1: The main term. One shows $\left\|\iota_q(k \cdot \iota_q^*\psi) - x\psi\right\|_{L^2} \le C'/q$ using the aliasing lemma (Lemma 4.1) to compare the sinc reconstruction of $k\cdot(\iota_q^*\psi)(k)$ with multiplication by x.

Step 2: The remainder. $\|\iota_q E_q\|_{L^2} \le (C/q)\left\|\iota_q(k^2 \iota_q^*\psi)\right\|_{L^2} \lesssim (C/q)\,\|x^2\psi\|_{L^2} = O(q^{-1})$ by the quasi-isometry bound (Corollary 4.2). □

6 Convergence of the momentum generator

Theorem 6.1 (Convergence of $\hat{P}_q$)**.** *For every $\psi \in \mathcal{S}(\mathbb{R})$,*

$$\left\| \iota_q\big(\hat{P}_q\, \iota_q^*\psi\big) - (-i\partial_x\psi) \right\|_{L^2(\mathbb{R})} = O\big(q^{-1}\big)\,,$$

uniformly on bounded subsets of $\mathcal{S}(\mathbb{R})$.

Proof sketch. The argument proceeds in three steps.

Step 1: Discrete algebraic decomposition (ℓ^2 level). By Lemma 3.1, $\hat{P}_q\, \iota_q^*\psi = g'_{\mathrm{disc}} + \frac{q}{4\pi i}(T_1 - 2I + T_{-1})\iota_q^*\psi$, where $g'_{\mathrm{disc}} = \frac{q}{4\pi i}(T_1 - T_{-1})\iota_q^*\psi$ is the centered discrete derivative. By Lemma 3.2 (which uses the aliasing lemma), the error term satisfies

$$\left\| \frac{q}{4\pi i}(T_1 - 2I + T_{-1})\iota_q^*\psi \right\|_{\ell^2} \le \frac{C}{q}\, \left\|\partial_x^2\psi\right\|_{L^2}\,.$$

Step 2: Approximation of the centered difference by the sampled continuous derivative. The centered difference $g'_{\mathrm{disc}}(k)$ approximates the *sampled continuous derivative* $(\partial_x\psi)|_{\mathbb{Z}/q\mathbb{Z}}$, i.e. the sequence $k \mapsto q^{1/2}(-i\partial_x\psi)(k/q)$ obtained via the adjoint embedding ι_q^* applied to $-i\partial_x\psi$, up to an aliasing error controlled by Lemma 4.1:

$$\left\| g'_{\mathrm{disc}} - (-i\partial_x\psi)|_{\mathbb{Z}/q\mathbb{Z}} \right\|_{\ell^2} \le \frac{C'}{q}\, \left\|\partial_x^3\psi\right\|_{L^2}\,.$$

The third-order derivative in the bound arises from the consistency error of the centered difference scheme: the leading term of the Taylor remainder for the centered approximation is of order $(\Delta x)^2\partial_x^3\psi \sim q^{-2}\partial_x^3\psi$, which after the ℓ^2-norm over q points and the factor q from the rescaling gives $O(q^{-1})\|\partial_x^3\psi\|_{L^2}$. Combining Steps 1 and 2, one obtains $\left\|\hat{P}_q\, \iota_q^*\psi - (-i\partial_x\psi)|_{\mathbb{Z}/q\mathbb{Z}}\right\|_{\ell^2} \le (C + C')/q \cdot \|\psi\|_{W^{3,2}}$.

Step 3: Passage from ℓ^2 to L^2 via aliasing. The aliasing lemma (Lemma 4.1) and the quasi-isometry (Corollary 4.2) give

$$\left\|\iota_q(\hat{P}_q\, \iota_q^*\psi) - (-i\partial_x\psi)\right\|_{L^2} \le \left\|\iota_q(\hat{P}_q\, \iota_q^*\psi - (\partial_x\psi)|_{\mathbb{Z}/q\mathbb{Z}})\right\|_{L^2} + \left\|\iota_q((\partial_x\psi)|_{\mathbb{Z}/q\mathbb{Z}}) - \partial_x\psi\right\|_{L^2}\,,$$

and both terms are bounded by $O(q^{-1})\,\|\partial_x^2\psi\|_{L^2}$. □

Remark 6.2. *The asymmetry between Theorems 5.1 and 6.1 is structural, not a proof artefact. The character $e^{2\pi ik/q}$ is a scalar multiplier; its Taylor expansion is local. The shift T_1 is a non-local operator; its approximation by a derivative requires a genuine discrete Sobolev estimate.*

7 Strong convergence on Schwartz data

Theorem 7.1 (Strong convergence on $\mathcal{S}(\mathbb{R})$)**.** *For every $\psi \in \mathcal{S}(\mathbb{R})$,*

$$\left\| \iota_q(\hat{X}_q\, \iota_q^*\psi) - x\psi \right\|_{L^2(\mathbb{R})} \xrightarrow{q\to\infty} 0, \qquad \left\| \iota_q(\hat{P}_q\, \iota_q^*\psi) + i\partial_x\psi \right\|_{L^2(\mathbb{R})} \xrightarrow{q\to\infty} 0.$$

The convergence is uniform on bounded subsets of $\mathcal{S}(\mathbb{R})$. This establishes Hypothesis [H2] *of [2].*

Proof. Immediate from Theorems 5.1 and 6.1. □

Remark 7.2 (Consequence for $Q5a$ Theorem 7.1)**.** *By [2] Theorem 7.1, together with the now-proved status of all six hypotheses ([H-Π1], [H-Π2]: proved in the O-series; [H-w]: proved in [8]; [H-E1], [C]: proved in [6]; [H1]: established on the admissible low-frequency sector; [H2]: established here), the full large-q limit theorem of $Q5a$ holds unconditionally on the admissible sector: the discrete oscillator representation of $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ converges to the metaplectic representation of $\mathrm{Heis}_3(\mathbb{R})$, and the admissibility forms converge in the Mosco sense to $\mathcal{E}(f,g) = A\int_{\mathbb{R}} \partial_x f\, \partial_x g\, dx$.*

8 Extension to $L^2(\mathbb{R})$

Theorem 8.1 (Extension to L^2)**.** *The strong convergence of Theorem 7.1 extends to all $\psi \in L^2(\mathbb{R})$: for every $\psi \in L^2(\mathbb{R})$ and $\varepsilon > 0$, there exists q_0 such that for all $q \geq q_0$,*

$$\left\| \iota_q(\hat{X}_q\, \iota_q^* \psi) - x\psi \right\|_{L^2} < \varepsilon, \qquad \left\| \iota_q(\hat{P}_q\, \iota_q^* \psi) + i\partial_x \psi \right\|_{L^2} < \varepsilon.$$

Proof sketch. Let $\psi_n \in \mathcal{S}(\mathbb{R})$ with $\psi_n \to \psi$ in $L^2(\mathbb{R})$. By Lemma 2.4, the operators $\iota_q \hat{X}_q \iota_q^*$ and $\iota_q \hat{P}_q \iota_q^*$ are bounded on the range of ι_q^* with norm $\leq q/\pi$. A standard 3ε argument using (i) the uniform bound of Lemma 2.4, (ii) the Schwartz convergence of Theorem 7.1 for ψ_n, and (iii) the density of $\mathcal{S}(\mathbb{R})$ in $L^2(\mathbb{R})$, yields the result. $\square$

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